

CHAPTER 15

Analog-to-Digital Converters

LEARNING OBJECTIVES

Upon completion of this chapter on analog-to-digital converters, you will be able to:

- Write the general output–input equations of an analog-to-digital converter (ADC) and calculate its output for any given input.
- Define and understand basic ADC specifications and terms.
- Name the most common types of ADCs and tell how each one operates.
- Explain how a microprocessor selects only one ADC out of all peripherals and reads its data.
- Calculate the maximum sine wave frequency that can be digitized to an accuracy of $\pm\frac{1}{2}$ LSB by an ADC or sample-and-hold amplifier.
- Operate a microprocessor-compatible ADC, the AD670.

15-0 INTRODUCTION

The previous chapter dealt with converting digital patterns of logic 1s and 0s to an analog signal—digital-to-analog converters (DACs). These devices are part of the output interface circuitry of a microprocessor-based system. They transform stored data or the results of digital processing back to real-world signals for control applications, for sound systems, or further analog processing, and so forth. In this chapter, we shall study part of a computer's input interface circuitry—*analog-to-digital converters*. ADCs are required to convert real-world signals produced by sensors and their signal-conditioning circuits to a digital pattern used by a computer. Digital data is easily stored, processed, and transmitted; hence, most modern control systems use some form of digital processing. Real-world parameters such as temperature, pressure, humidity, and so on are first converted by a sensor to an electrical quantity and then signal-conditioned analog circuitry before being applied to an A/D converter. These analog signals may vary slowly with time (for example, temperature measurements) or quickly with time (for example, audio recording). The latter systems require sample-and-hold circuitry before the signal can be applied to an A/D converter.

15-1 ADC CHARACTERISTICS

15-1.1 Resolution

The digital output of an ideal 3-bit ADC is plotted against a normalized analog input in Fig. 15-1. Similar to a DAC, the resolution of an ADC is defined as

$$\text{resolution} = 2^n \quad (15-1)$$

where n is the number of digital outputs. Since the input is an analog voltage and assumed to be continuous, resolution refers to number of discrete outputs. The ratio of the device's full-scale input voltage range, FSR, to the resolution gives the minimum change of input voltage to cause a change of one digital count at the output. In equation form

$$\Delta V_i \text{ for 1 LSB} = \frac{\text{FSR}}{2^n} \quad (15-2)$$

where FSR is the full-scale input voltage range specified on the manufacturer's data sheet and n is the number of digital outputs. The ratio of Eq. (15-2) is simply referred to as 1 LSB as shown in Fig. 15-1. *Note:* For an A/D converter 1 LSB is a voltage value as we will see in Example 15-1.

Since the digital count begins at 0, then the maximum full-scale input voltage to cause the output to be all logic 1s is 1 LSB less than the full-scale range value.

$$V_{\text{ifs}} = \text{FSR} - 1 \text{ LSB} \quad (15-3)$$

where V_{ifs} is the maximum input voltage to cause all logic 1s at the ADC's output. In general, the output–input equation of an ADC for any input voltage is given by

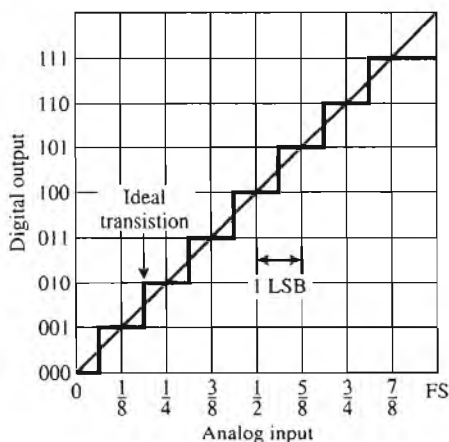


FIGURE 15-1 Ideal input-output relationship for a 3-bit A/D converter.

$$\text{digital output code} = \text{binary equivalent of } D \quad (15-4)$$

where D is the number of LSBs and is found from

$$D = \frac{V_{in}}{1 \text{ LSB}} \quad (15-5)$$

V_{in} is the applied input voltage and 1 LSB is given by Eq. (15-2).

Example 15-1

An 8-bit ADC is capable of accepting an input unipolar (positive values only) voltage 0 to 10 V. (a) What is the minimum value of input voltage to cause a digital output change of 1 LSB? (b) What input voltage will cause all 1s at the ADC's output? (c) What is the digital output code if the applied input voltage is 5.2 V?

Solution (a) From Eq. (15-2)

$$1 \text{ LSB} = \frac{10 \text{ V}}{2^8} = 39.1 \text{ mV}$$

(b) From Eq. (15-3)

$$V_{ifs} = 10 \text{ V} - 39.1 \text{ mV} = 9.961 \text{ V}$$

(c) Applying Eq. (15-5)

$$D = \frac{5.2 \text{ V}}{39.1 \text{ mV}} = 132.99 \approx 133$$

Converting this value to binary yields 10000101.

Figure 15-1 shows the graph for an ideal 3-bit A/D converter. Since there are only 8 discrete outputs for this ADC, the analog input range from 0 to full scale must be *quantized* into these 8 digital outputs. Hence, voltages within the same range have the same digital code. The ideal transition from one digital code to the next is taken to be at the mid-range value as shown. A/D converters like DACs, have offset, gain, and linearity errors but they also have a quantization error.

15-1.2 Quantization Error

Figure 15-1 shows that the digital output is 100 for all values of $\frac{1}{2} \text{FSR} \pm \frac{1}{2} \text{LSB}$. Hence, there is an unavoidable uncertainty about the exact value of V_{in} when the output is 100. This uncertainty is specified as quantization error. Its value is $\pm \frac{1}{2} \text{LSB}$. Increasing the number of bits results in a finer resolution and therefore there is a smaller quantization error.

15-1.3 Offset Error

Similar to DACs, analog-to-digital converters also have an offset error specification. It is usually given as a fraction of 1 LSB such as $\pm \frac{1}{2} \text{LSB}$. Figure 15-2 shows how an A/D

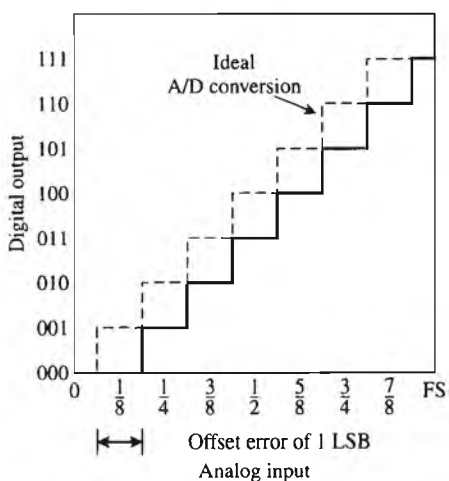


FIGURE 15-2 Comparison of the ideal A/D conversion with a conversion having offset error.

converter's output–input plot can be viewed with an offset error. This offset error modifies Eq. (15-3) as shown in the next example.

Example 15-2

Consider that the 8-bit A/D converter of Problem 15-1 has an offset error of $\pm \frac{1}{2}$ LSB. What analog input voltage will cause a digital output code of all 1s?

Solution From the solution of Problem 15-1, 1 LSB is 39.1 mV. Therefore, $\frac{1}{2}$ LSB is 19.5 mV. Modifying Eq. (15-3) by the offset error yields

$$\begin{aligned} V_{\text{ifs}} &= \text{FSR} - 1 \text{ LSB} + \frac{1}{2} \text{ LSB} \\ &= 10 \text{ V} - 39.1 \text{ mV} + 19.5 \text{ mV} = 9.9804 \text{ V} \end{aligned}$$

15-1.4 Gain Error

Refer to Fig. 15-1. In the ideal input–output relationship the difference between the first transition and the last transition is $\text{FSR} - 2 \text{ LSB}$. If this expression is not true, there is a gain error. See Fig. 15-3 to compare the ideal relationship with an A/D converter that has a gain error. Manufacturers specify gain error as a percentage of full-scale voltage (% FSR).

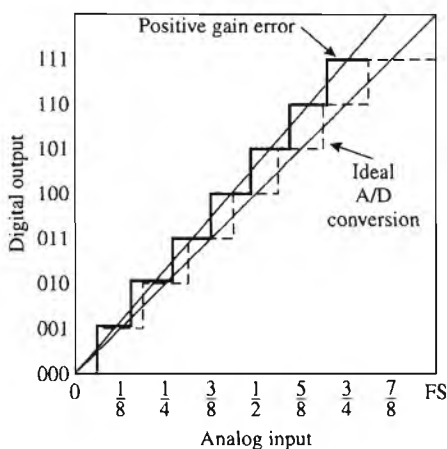


FIGURE 15-3 Gain error.

Example 15-3

The A/D converter of Problem 15-1 has a gain error of 0.5% of FSR. What value of V_{in} will cause the output digital code to be all 1s?

Solution In terms of voltage, the gain error is 50 mV ($0.5\% \times 10 \text{ V}$). The ideal full-scale input voltage to cause the output to be all logic 1s is given by Eq. (15-3) and from Problem 15-1 part (c) V_{ifs} is 9.961 V. This value is reduced by a positive gain error. Therefore the new value of input voltage is $V_{in} = 9.961 \text{ V} - 50 \text{ mV} = 9.911 \text{ V}$.

15-1.5 Linearity Error

Linearity error in an A/D converter exists if the differences between all of the transitions are not equal. Figure 15-4 shows that the difference between transitions are not equal. The range difference is specified as a fraction of 1 LSB such as 0.75 LSB.

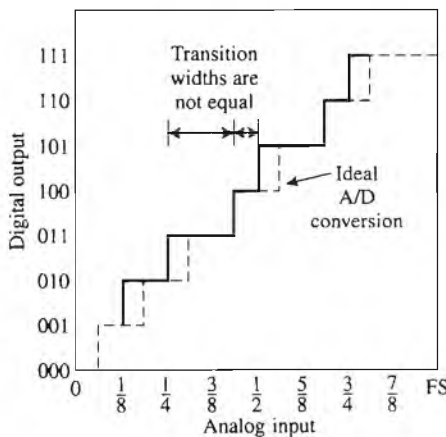


FIGURE 15-4 Linearity error.

15-2 INTEGRATING ADC**15-2.1 Types of ADCs**

General characteristics of ADCs were introduced in Section 15-1. There are three standard types classified according to their conversion times. The *slow integrating ADC* typically requires 300 ms to perform a conversion; it is the best choice for measuring slowly

varying dc voltages. The faster *successive approximation* ADC's conversion time is a few microseconds and can digitize audio signals. Fastest of all are the more costly *flash* converters, which can digitize video signals.

15-2.2 Principles of Operation

The block diagram of a typical dual-slope integrating A/D converter is shown in Fig. 15-5. An on-board divide-by-4 counter drives the control logic at a rate of 12 kHz. This frequency is set by the user via external timing resistor R_T and C_T . It must be a multiple of the local line frequency (50 or 60 Hz) to render the ADC immune to line-frequency noise.

The control logic unit activates a complex network of logic circuits and analog switches to convert analog input voltage V_{in} into a digital output. The analog-to-digital conversion is performed in three phases and requires about one-third of a second. These operating phases are called *signal integrate phase* T_1 , *reference integrate phase* T_2 , and *auto-zero phase* T_z . These will be discussed in sequence.

15-2.3 Signal Integrate Phase, T_1

The control logic unit of Fig. 15-5(b) connects V_{in} to an integrator to begin phase T_1 . The integrator or ramp generator's output V_o ramps up or down depending on the polarity of V_{in} and at a rate set by V_{in} , R_{int} , and C_{int} . If V_{in} is negative, V_o ramps up, as shown in Fig. 15-5(a). Time T_1 is set by the logic unit for 1000 clock pulses. Since the 12-kHz clock has a period of 83.3 μ s per count, T_1 lasts 83.33 ms.

If $V_{in} = -100$ mV, V_o will ramp from 0 V to 833 mV. The maximum allowed full-scale value of V_{in} is ± 200 mV. When $V_{in} = -200$ mV, V_o will rise to a maximum 1666 mV. Clearly, V_o is directly proportional to V_{in} . At the end of 1000 counts, the logic unit disconnects V_{in} and connects V_{ref} to the integrator. This action ends T_1 and begins T_2 .

15-2.4 Reference Integrate Phase, T_2

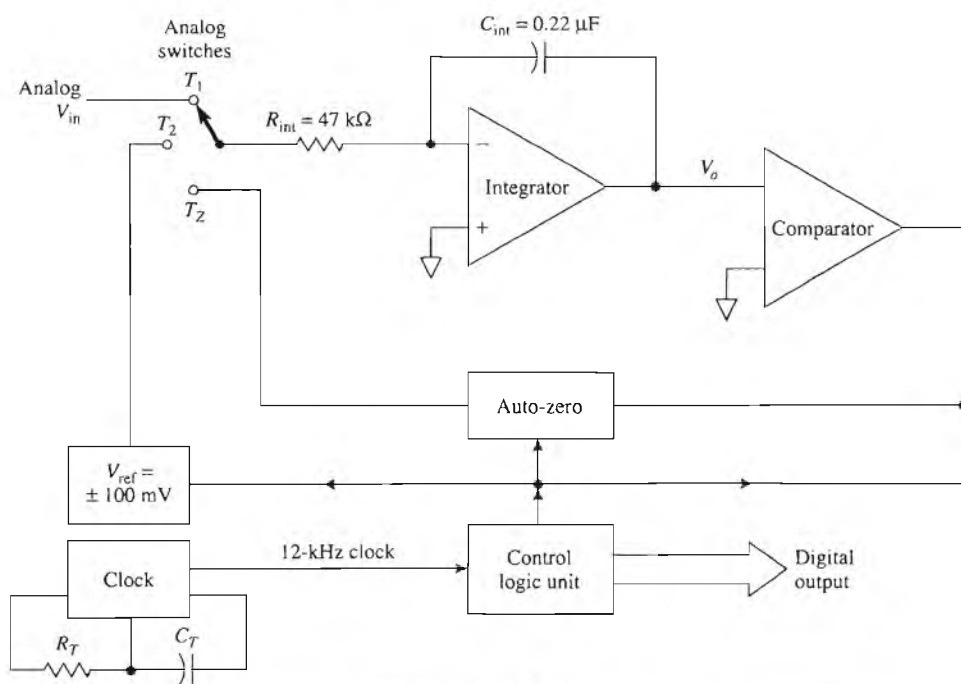
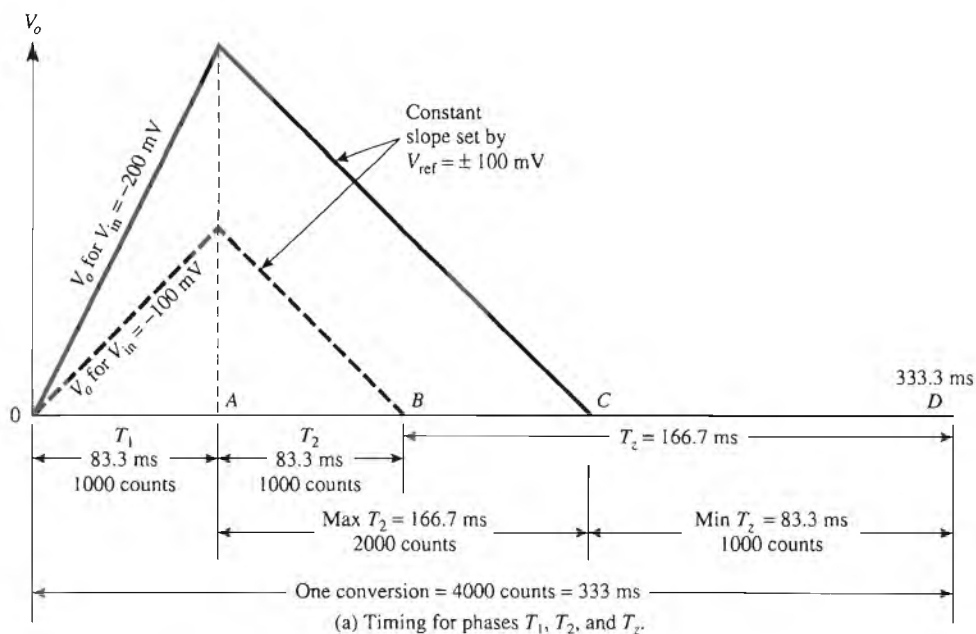
During phase T_1 , the logic unit determined the polarity of V_{in} and charged a reference capacitor, C_{ref} (not shown), to a reference voltage $V_{ref} = 100$ mV. At the beginning of phase T_2 , the logic unit connects C_{ref} to the integrator so that V_{ref} has a polarity opposite to V_{in} . Consequently, V_{ref} will ramp the integrator back toward zero. Since V_{ref} is constant, the integrator's output V_o will ramp down at a *constant* rate, as shown in Fig. 15-5a.

When V_o reaches zero, a comparator "tells" the logic unit to terminate phase T_2 and begin the next auto-zero phase. T_z is thus proportional to V_o and consequently, V_{in} . The exact relationship is

$$T_2 = T_1 \frac{V_{in}}{V_{ref}} \quad (15-6a)$$

Since $T_1 = 83.33$ ms and $V_{ref} = 100$ mV,

$$T_2 = \left(0.833 \frac{\text{ms}}{\text{mV}} \right) V_{in} \quad (15-6b)$$



(b) Simplified block diagram.

FIGURE 15-5 (a) Timing diagram of a typical dual-slope integrating ADC; (b) simplified block diagram of a dual-slope integrating ADC. In (a), one A/D conversion takes place in three phases: signal integrating phase T_1 , reference T_2 , and auto-zero T_z .

Example 15-4

For the ADC of Fig. 15-5, calculate T_2 if (a) $V_{in} = \pm 100$ mV; (b) $V_{in} = \pm 200$ mV.

Solution (a) From Eq. (15-6b),

$$T_2 = \left(0.833 \frac{\text{ms}}{\text{mV}} \right) (100 \text{ mV}) = 83.33 \text{ ms}$$

(b)

$$T_2 = \left(0.833 \frac{\text{ms}}{\text{mV}} \right) (200 \text{ mV}) = 166.6 \text{ ms}$$

15-2.5 The Conversion

The actual conversion of analog voltage V_{in} into a digital count occurs during T_2 as follows. The control unit connects the clock to an internal binary-coded-decimal counter at the beginning of phase T_2 . The clock is disconnected from the counter at the end of T_2 . Thus the counter's content becomes the digital output. This digital output is set by T_2 and the clock frequency:

$$\text{digital output} = \left(\frac{\text{counts}}{\text{second}} \right) T_2 \quad (15-7a)$$

but T_2 is set by V_{in} from Eq. (15-6a) and therefore

$$\text{digital output} = \left(\frac{\text{counts}}{\text{second}} \right) (T_1) \left(\frac{V_{in}}{V_{ref}} \right) \quad (15-7b)$$

Since clock frequency is 12 kHz for the 7106/7107 ADC, $T_1 = 83.33$ ms, and $V_{ref} = 100$ mV, the output-input equation is

$$\text{digital output} = \left(12,000 \frac{\text{counts}}{\text{second}} \right) \left(\frac{83.33 \text{ ms}}{100 \text{ mV}} \right) V_{in}$$

or

$$\text{digital output} = \left(10 \frac{\text{counts}}{\text{mV}} \right) V_{in} \quad (15-7c)$$

The counter's output is connected to an appropriate $3\frac{1}{2}$ -digit display.

Example 15-5

V_{in} equals + 100 mV in the ADC of Fig. 15-5. Find the digital output.

Solution From Eq. (15-7c),

$$\text{digital output} = \left(10 \frac{\text{counts}}{\text{mV}} \right) (100 \text{ mV}) = 1000 \text{ counts}$$

Example 15-5 shows the need for some human engineering. The display reads 1000, but it *means* that V_{in} equals 100 mV. *You* must wire in a decimal point to display 100.0 and paste an “mV” sign beside the display.

15-2.6 Auto-Zero

The block diagram of Fig. 15-5(b) contains a section labeled “Auto-zero.” During the third and final phase of conversion, T_z , the logic unit activates several analog switches and connects an auto-zero capacitor C_{AZ} (not shown).

The auto-zero capacitor is connected across the integrating capacitor, C_{int} , and any input offset voltages of both integrating and comparator op amps. C_{AZ} charges to a voltage approximately equal to the average error voltage due to C_{int} and the offset voltages. During the following phases, T_1 and T_2 , the error voltage stored on C_{AZ} is connected to cancel any error voltage on C_{ref} . Thus the ADC is automatically zeroed for every conversion.

15-2.7 Summary

Refer to the timing diagram in Fig. 15-5(a). The logic unit allocates 4000 counts for one conversion. At 83.33 μs per count, the conversion takes 333 ms. The control unit always allocates 1000 counts or 83.3 ms to phase T_1 .

The number of counts required for T_2 depends on V_{in} . Zero counts are used for $V_{in} = 0 \text{ V}$ and a maximum of 2000 counts, or 166.7 ms are used when V_{in} is at its maximum limit of $\pm 200 \text{ mV}$.

T_2 and T_z always share a total of 3000 counts for a total of 250 ms. For $V_{in} = 0 \text{ V}$, $T_2 = 0$ counts and $T_z = 3000$ counts. For $V_{in} = \pm 200 \text{ mV}$, $T_2 = 2000$ counts and $T_z = 1000$ counts.

Some manufacturers of ADCs market a complete $3\frac{1}{2}$ -digit digital voltmeter kit. The kit contains a 40-pin dual-slope integrating A/D converter, all necessary parts, printed circuit board, and instructions. The instructions make it easy to construct and easy to use, and form an excellent tutorial on integrating ADCs.

15-3 SUCCESSIVE APPROXIMATION ADC

The block diagram of a successive approximation register (ADC) is shown in Fig. 15-6. It consists of a DAC, a comparator, and a *successive approximation register* (SAR). One terminal is required for analog input voltage V_{in} . The digital output is available in either serial or parallel form. A minimum of three control terminals are required. *Start conversion* initiates an A/D conversion sequence and *end of conversion* tells when the conversion is completed. An external clock terminal sets the time to complete each conversion.

15-3.1 Circuit Operation

Refer to Fig. 15-6. An input *start conversion* command initiates one analog-to-digital conversion cycle. The successive approximation register (SAR) connects a sequence of digital numbers, one number for each bit to the inputs of a DAC. This process was explained in Chapter 14.

The DAC converts each digital number into an analog output V_o . Analog input voltage, V_{in} , is compared to V_o by a comparator. The comparator tells the SAR whether V_{in} is greater or less than DAC output V_o , once for each bit. For a 3-bit output, three comparisons would be made.

Comparisons are made beginning with the MSB and ending with the LSB, as will be explained. At the end of the LSB comparison, the SAR sends an end-of-conversion signal. The digital equivalent of V_{in} is now present at the SAR's digital output.

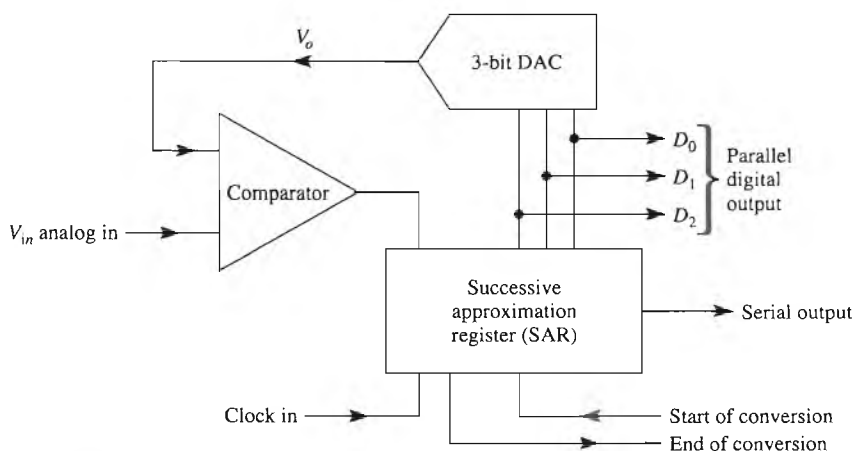


FIGURE 15-6 Block diagram of a successive approximation 3-bit ADC.

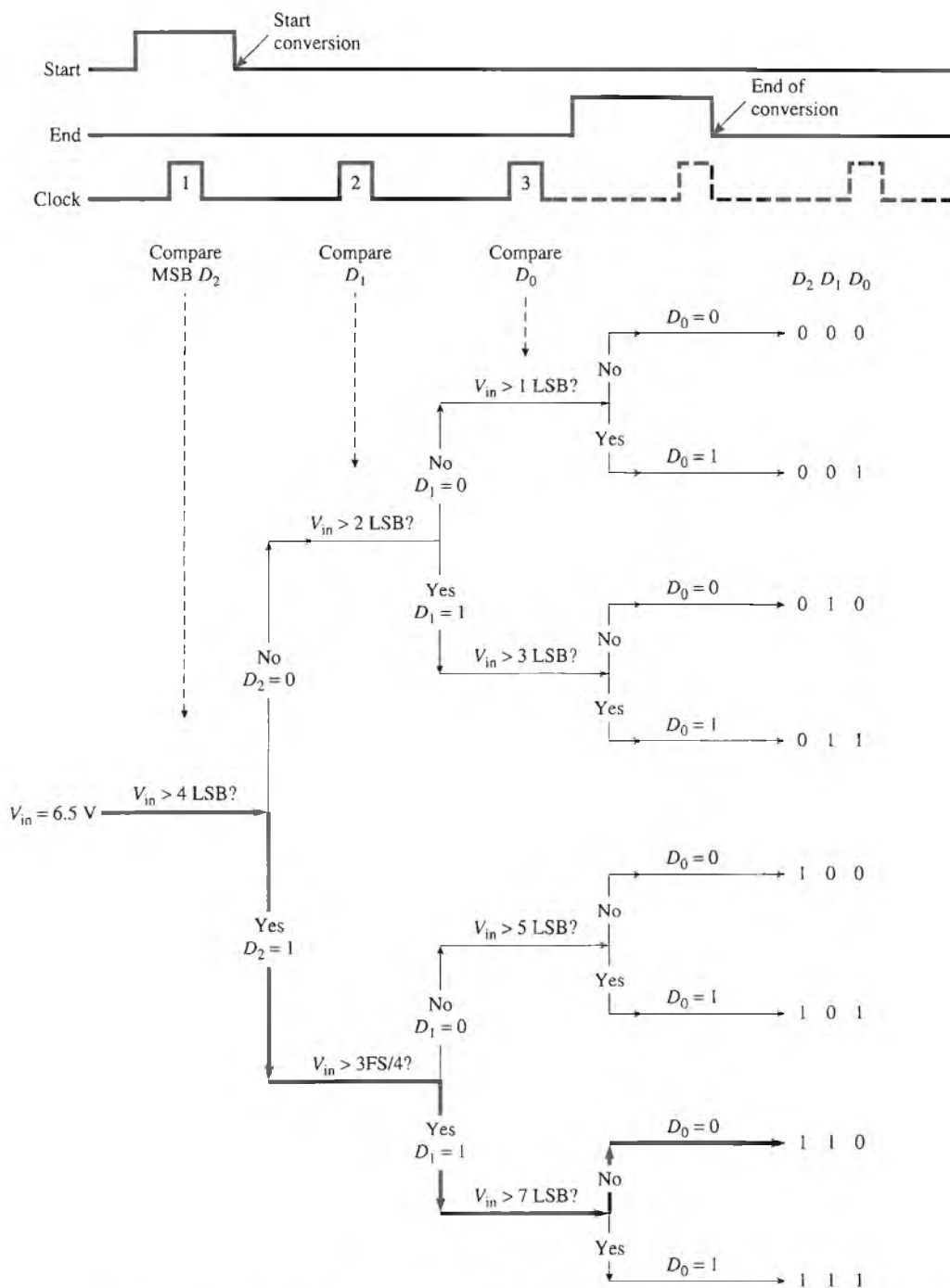


FIGURE 15-7 “Start” begins operation of this 3-bit successive approximation register. Beginning with the MSB, the weight of each bit is compared with V_{in} by the comparator of Fig. 15-6. If V_{in} is greater, the SAR’s output is set to 1, or to 0 if V_{in} is smaller. Dark lines show the conversion for $V_{in} = 6.5 \text{ V}$.

15-3.2 Successive Approximation Analogy

Suppose that you had 1-, 2-, and 4-lb weights (SAR) plus a balance scale (comparator and DAC). Think of the 1-lb weight as 1 LSB and the most significant 4-lb weight as 4 LSB. Refer to Figs. 15-6 and 15-7. V_{in} corresponds to an unknown weight.

Let us convert $V_{in} = 6.5$ V to a digital output (unknown weight = 6.5 lb). You would place the unknown weight on one platform of the balance, the 4-lb weight on the other, and compare if the unknown weight (V_{in}) exceeded the 4-lb weight. The SAR uses one clock pulse to apply the MSB 100 to the DAC in Fig. 15-7. Its output, $V_o = 4$ V, is compared with V_{in} . The MSB (D_2) is set to 1 if $V_{in} > V_o$. This is analogous to your leaving the 4-lb weight on the scale.

The SAR then applies 110 (add a 2-lb weight) to the DAC, D_1 is set to 1 since $V_{in} = 6.5$ V is greater than $V_o = 6$ V. Finally, the SAR applies 111 to the DAC (add 1 lb). Since $V_{in} = 6.5$ V is less than 7 V, D_0 is set to zero (1-lb weight removed).

15-3.3 Conversion Time

Figure 15-7 shows that one clock pulse is required for the SAR to compare each bit. However, an additional clock pulse is usually required to reset the SAR prior to performing a conversion. The time for one analog-to-digital conversion must depend on both the clock's period T and number of bits n . The relationship is

$$T_C = T(n + 1) \quad (15-8)$$

Example 15-6

An 8-bit successive approximation ADC is driven by a 1-MHz clock. Find its conversion time.

Solution The time for one clock pulse is 1 μ s. From Eq. (15-8),

$$T_C = 1 \mu\text{s}(8 + 1) = 9 \mu\text{s}$$

15-4 ADCs FOR MICROPROCESSORS

One of the principal selling points of many microcontrollers is an internal A/D converter. However, microprocessors do not have internal A/Ds, so we must use an external device for analog-to-digital conversion. The microprocessor “views” a peripheral ADC simply as a “read only” address in the microprocessor’s memory map. Refer to Fig. 15-8. The ADC must have a tristate *memory buffer register* (MBR). In the idle state, the MBR will contain a digital code resulting from the ADC’s *last* conversion. Also, the MBR will be disconnected from the data bus.

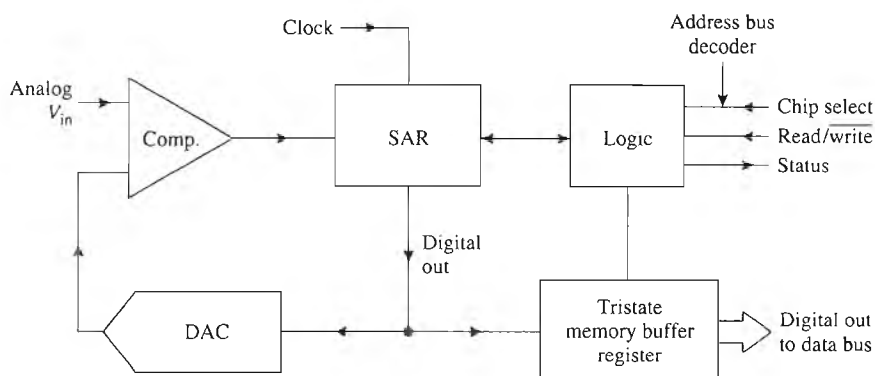


FIGURE 15-8 To be compatible with microprocessors, the ADC of Fig. 15-6 requires selection logic and a memory buffer register.

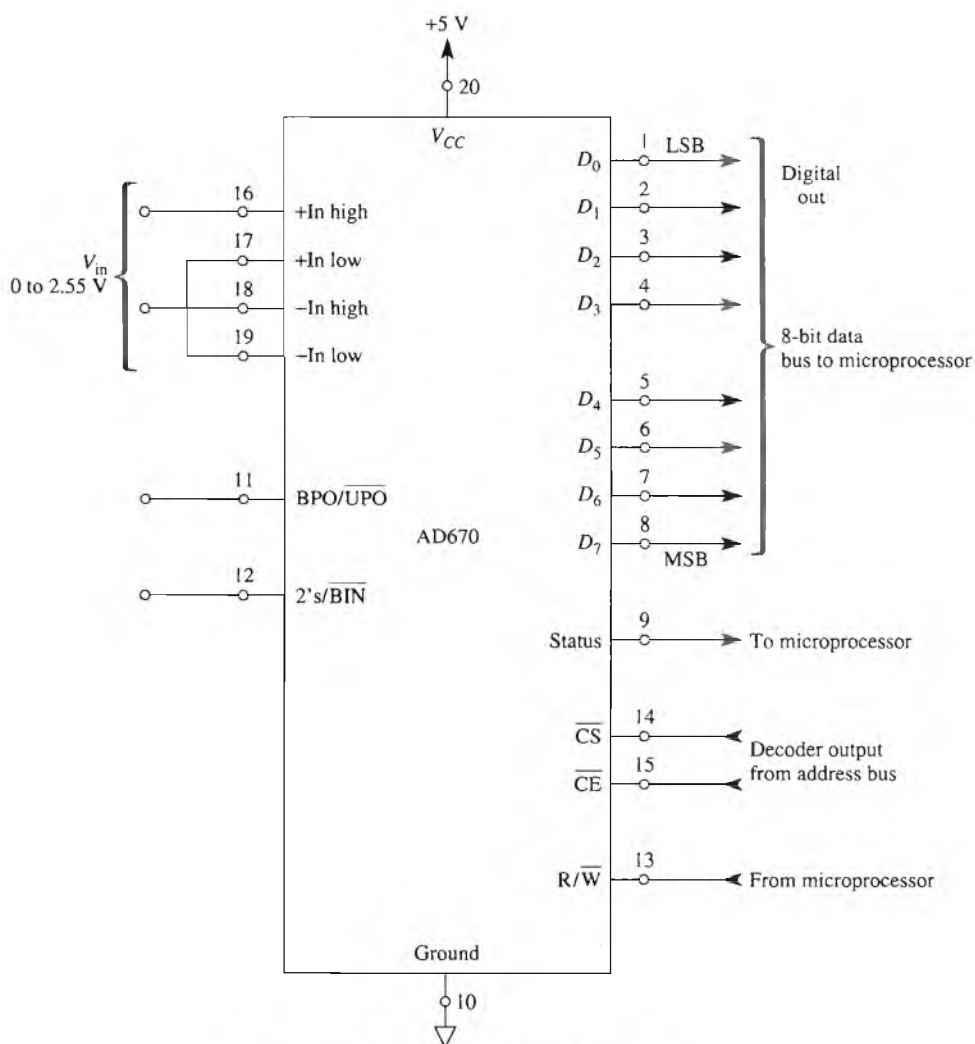
The microprocessor uses the address bus and decoders to select one ADC out of all the others by bringing its *chip select* terminal low. A low on the *chip select* terminal in Fig. 15-8 tells the ADC that a command is coming to its read/write terminal. If read/write is brought low by the microprocessor, the ADC converts V_{in} into a digital code and loads or writes it into its own MBR. When read/write is high and chip select is low, the ADC's memory buffer register is connected (transparent) to the data bus.

It is important to look at this operation from the microprocessor's viewpoint. A *read* command means that the microprocessor is going to read data stored in the ADC's memory buffer register. The ADC's digital tristate outputs must go from high-Z (high impedance) to transparent and connect the digital word to the data bus. A *write command* is actually a *start conversion command* to the ADC. The microprocessor thus tells the ADC: (1) perform a conversion; (2) store (and *write*) it in your memory; and (3) don't tell me the result until I want to *read* it.

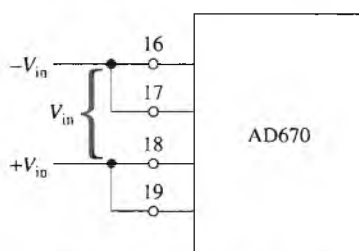
Finally, the microprocessor-compatible ADC must tell the microprocessor via its *status* line when a conversion is in progress; the status line goes high. If a conversion is completed, the status line goes low to signal the microprocessor that data is valid and ready for reading. We select the Analog Devices AD670 to learn how all the foregoing features are available in a single 20-pin integrated circuit.

15-5 AD670 MICROPROCESSOR-COMPATIBLE ADC

The AD670 is an 8-bit microprocessor-compatible successive approximation analog-to-digital converter. The 20-pin package of Fig. 15-9 contains all the features described in Section 15-3 and Fig. 15-8. In addition, it contains an on-board clock, voltage reference, and instrumentation amplifier, and needs only a single 5-V supply. To understand how the AD670 operates, we examine the tasks performed by each of its terminals and associated circuit blocks.



(a) Inputs wired for a resolution of 10 mV/bit.



(b) Inputs wired for 1 mV/bit.

FIGURE 15-9 (a) AD670 ADC pin connections. Full-scale analog input voltages are 0 to 2.55 V or 0 to ± 1.28 V and 0 to 255 mV or 0 to ± 128 mV in (b).

15-5.1 Analog Input Voltage Terminals

Four analog input terminals are pins 16, 17, 18, and 19 in Fig. 15-9. They are inputs to an instrumentation amplifier configured to handle unipolar or bipolar analog input voltages. They are also pin-programmable to make it easy for the user to select resolution. Figure 15-9(a) shows operation for an analog input of 0 to 2.55 V, resolution = 10 mV/LSB. Figure 15-9(b) shows operation for 0 to 255 mV or 1 mV/LSB.

15-5.2 Digital Output Terminals

Pins 1 through 8 are tristate, buffered, latching digital outputs for the data bus digits, D_0 through D_7 , respectively. When a microprocessor tells the AD670 to perform a conversion (write), the result is latched into its memory buffer register. Tristate output switches are held in the *high-impedance* (high- Z) state until the microprocessor sends a read command. Thus the ADC's memory register is normally disconnected from the data bus.

15-5.3 Input Option Terminal

Pin 11 is called $\overline{\text{BPO/UPO}}$ and allows the microprocessor to tell the AD670 whether to accept a bipolar analog input voltage range or a unipolar input range. A low on pin 11 selects unipolar operation. A range of 0 to 2.55 V or 0 to 255 mV is set by the user as in Figs. 15-9(a) and (b). A high sent to pin 11 selects bipolar operation. The V_{in} range is then ± 1.28 V [Fig. 15-9(a)] or ± 128 mV [Fig. 15-9(b)].

15-5.4 Output Option Terminal

In Fig. 15-9, pin 12 is labeled "2's/ $\overline{\text{BIN}}$." It allows the microprocessor to tell the AD670 to present an *output format* in either 2's-complement code or binary code. A binary output code format will be *straight binary* if V_{in} is unipolar (pin 11 = low) or *offset binary* if V_{in} is bipolar (pin 11 = high). The four possible options are shown in Fig. 15-10(a).

The digital output responses to analog input V_{in} are shown in Figs. 15-10(b) and (c). V_{in} is the *differential* input voltage and is defined by

$$V_{\text{in}} = (+V_{\text{in}}) - (-V_{\text{in}}) \quad (15-9)$$

where $+V_{\text{in}}$ and $-V_{\text{in}}$ are measured with respect to ground.

15-5.5 Microprocessor Control Terminals

As shown in Fig. 15-9, pins 13, 14, and 15 are used by a microprocessor to control the AD670. Terminal 14 is called *chip select* ($\overline{\text{CS}}$) and terminal 15 is called *chip enable* ($\overline{\text{CE}}$). Pin 13 is called *read/write* ($\overline{\text{R/W}}$).

If $\overline{\text{CS}}$, $\overline{\text{CE}}$, and $\overline{\text{R/W}}$ are all brought low, the ADC converts continuously. It performs one conversion every 10 μs or less. The result of each conversion is latched into the output buffer register. However, the digital output code is *not* connected to the data bus

Pin 11 BPO/UPO	Input range	Pin 12 2's/BIN	Output format
0	Unipolar	0	Straight binary
1	Bipolar	0	Offset binary
0	Unipolar	1	2's complement
1	Bipolar	1	2's complement

(a)

Differential V_{in}	Unipolar/straight binary, pin 11 = 0, 12 = 0
0	0000 0000
1 mV	0000 0001
128 mV	1000 0000
255 mV	1111 1111

(b)

Differential V_{in}	Bipolar/offset binary, pin 11 = 1, 12 = 0	Bipolar/2's complement, pin 11 = 1, 12 = 1
-128 mV	0000 0000	1000 0000
-1 mV	0111 1111	1111 1111
0	1000 0000	0000 0000
1 mV	1000 0001	0000 0001
127 mV	1111 1111	0111 1111

(c)

FIGURE 15-10 Input ranges are unipolar or bipolar, and output formats are determined by pins 11 and 12 in (a). Output codes are given for unipolar inputs in (b) and for bipolar inputs in (c). (a) Input range and output format are controlled by pins 12 and 11, respectively; (b) digital output codes for unipolar V_{in} inputs wired as in Fig. 15-9a; (c) digital output codes for bipolar V_{in} inputs wired as in Fig. 15-9b.

because the outputs are high impedance. This condition is called a *write and convert* command; that is, the microprocessor tells the AD670 to write converted data into its own buffer register. If $\overline{CS}$ or $R/\overline{W}$ or $\overline{CE}$ is high, the AD670 is unselected (high impedance) and retains the last conversion in its register.

The status terminal, pin 9, stays high during a conversion. When a conversion is completed, pin 9 outputs a low to “tell” the microprocessor that data is valid in the AD670’s buffer register. To read data out of the AD670, the microprocessor brings $R/\overline{W}$ high while status and $\overline{CS}$ and $\overline{CE}$ are low. This is a *read* command from the microprocessor.

The AD670’s buffer becomes transparent and connects the eight digital outputs (D_7 through D_0) to the data bus. Data will remain on the bus until the AD670 is disconnected by bringing $\overline{CS}$ high, $\overline{CE}$ high, or $R/\overline{W}$ low.

Summary

1. A low on $\overline{CE}$ and $\overline{CS}$ selects the AD670. What happens next depends on $R/\overline{W}$.
2. If $R/\overline{W}$ is low (for at least $0.3 \mu s$), a conversion is performed and the result is written into the buffer register. Outputs are high impedance. The conversion requires $10 \mu s$.

3. If $\overline{R/W}$ is high, the last conversion is stored in the buffer and the outputs are transparent. No further conversions are performed. The contents of the register can now be read by the microprocessor via the data bus.
4. Status tells the microprocessor what is going on within the AD670. Status = high means that conversion is being performed. Status = low tells the microprocessor that data are valid. The microprocessor is free to read the selected AD670's data by placing a high to $\overline{R/W}$.

15-6 TESTING THE AD670

Figure 15-11 shows how to wire an AD670 to perform continuous conversions *without* a microprocessor. This circuit can be used as a laboratory exercise to gain experience operating ADCs. Each data output, D_0 through D_7 , is connected to an inverter, resistor, and LED. These components simulate a data bus. An LED lights to signify that a logic 1 is present on its associated data bus wire.

Pins 14 and 15 are wired so that $\overline{CS}$ and $\overline{CE}$ are low. This causes continuous conversion. The 555 timer drives $\overline{R/W}$ low for 5 μs to simulate a write command. $\overline{R/W}$ thus returns high before a conversion is completed in the 10- μs conversion time. At the end of 10 μs , the high on $\overline{R/W}$ simulates a read command and data are displayed on the LEDs. If $R_T = 1.5 \text{ M}\Omega$, the AD670 makes one conversion and one readout 1000 times per second. Reduce R_T to 120 k Ω for convert/reads of 10,000 times per second.

15-7 FLASH CONVERTERS

15-7.1 Principles of Operation

Fastest of all A/D converters is the *flash* converter, shown in Fig. 15-12(a). A reference voltage and resistor divider network establishes a resolution of 1 V/LSB. Analog input voltage V_{in} is applied to the + inputs of all comparators. Their outputs drive an 8-line-to-3-line priority encoder. The encoder logic outputs a binary code that represents the analog input.

For example, suppose that $V_{in} = 5.0 \text{ V}$. The outputs of comparators 1 through 5 would go high and 6 through 8 would go low. As shown in Fig. 15-12(b), the digital output would be 101.

15-7.2 Conversion Time

The conversion time of the flash converter is limited only by the response times of comparators and logic gates. They can digitize video or radar signals. The flash converter's high speed becomes more expensive as resolution is increased. Figure 15-12 shows that the flash converter requires seven comparators (or $2^3 - 1$) to perform a 3-bit conversion. The number of comparators required for n -bit resolution is

$$\text{number of comparators} = 2^n - 1 \quad (15-10)$$

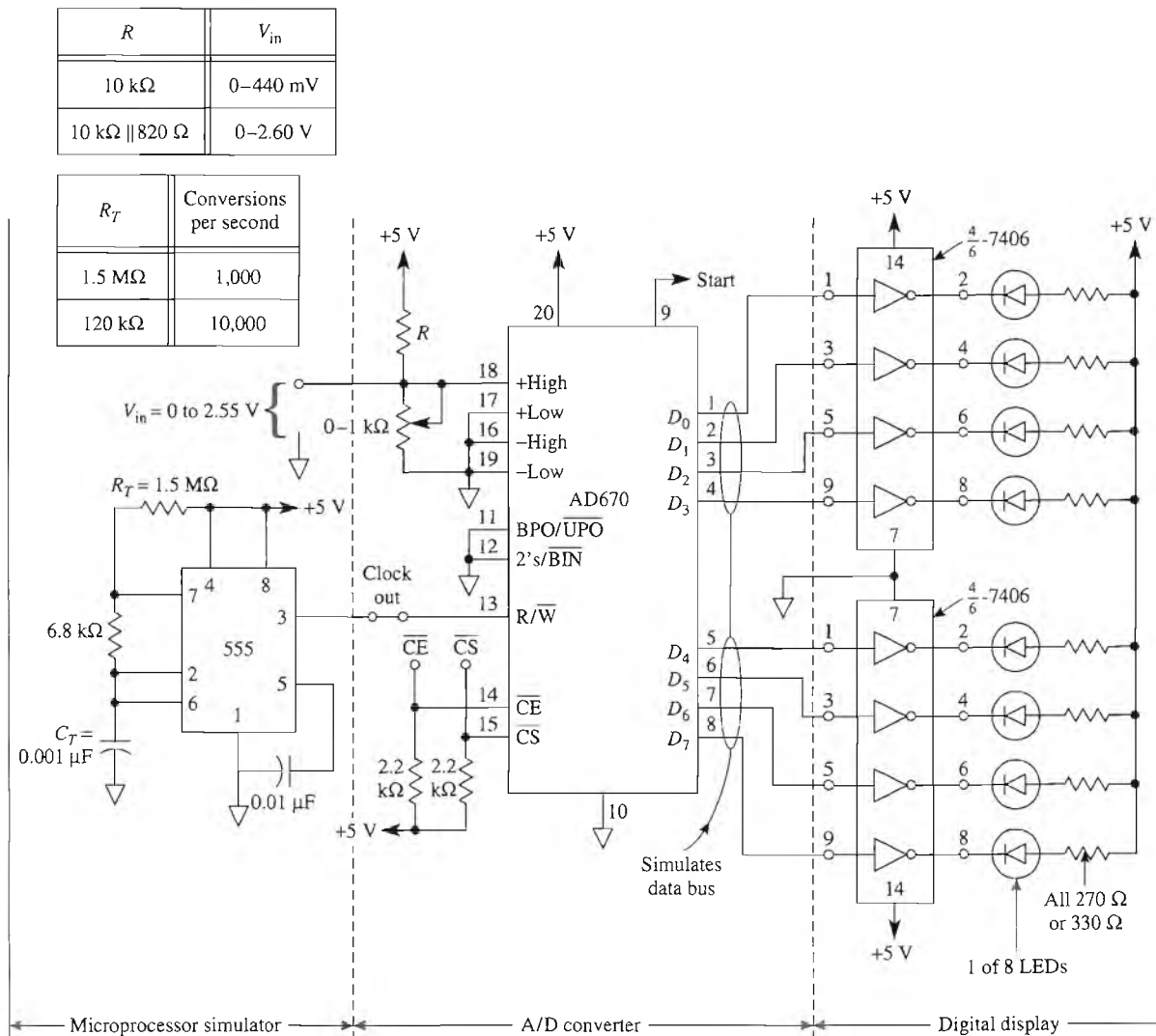


FIGURE 15-11 The operation of the AD670 can be studied without the need for a microprocessor. Pins 14 and 15 can be grounded to simulate a microprocessor selection via an address bus. The 555 timer simulates continuous read/write commands from a micro-

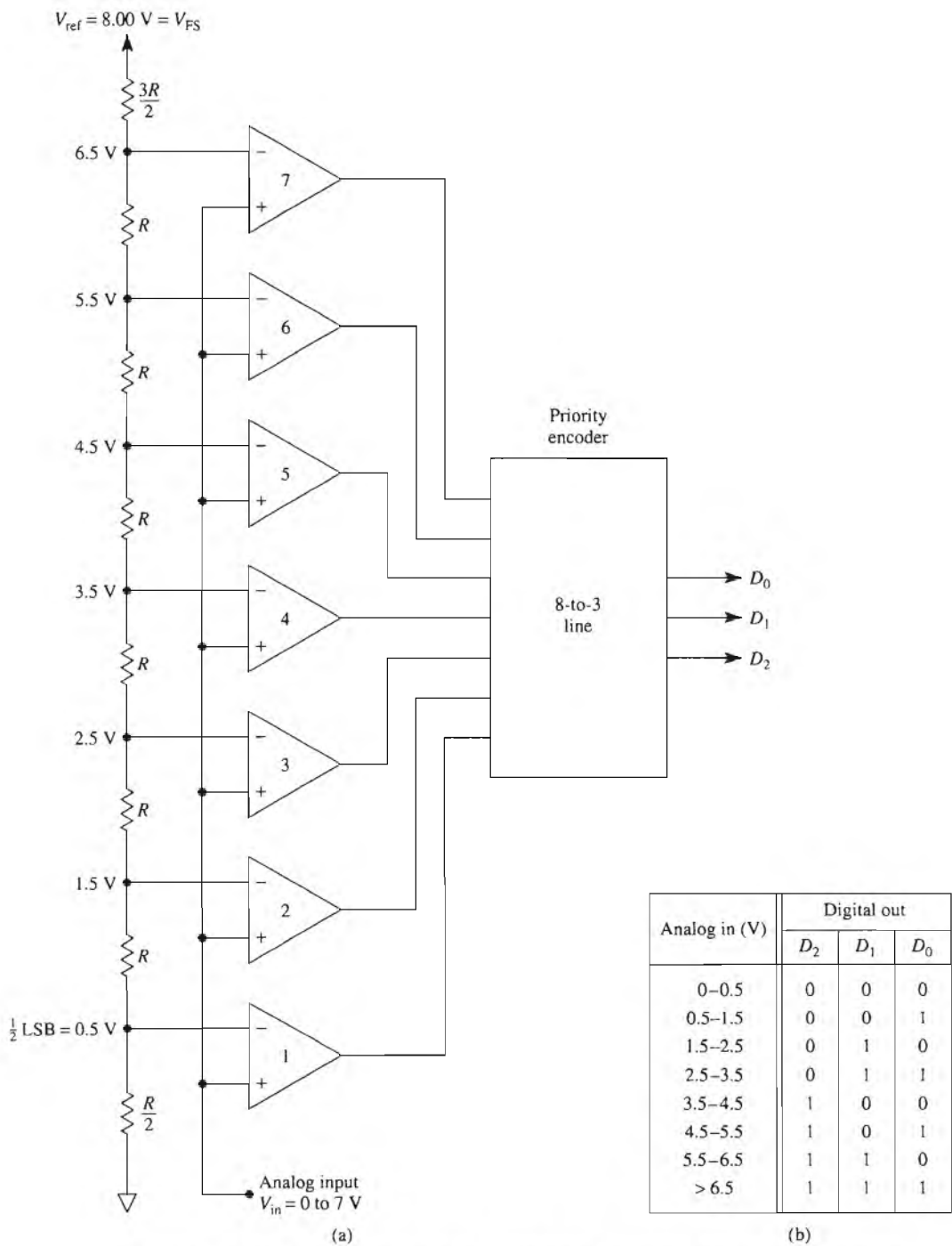


FIGURE 15-12 (a) Three-bit flash (parallel) A/D converter; (b) output versus input.

For example, an 8-bit flash converter requires $(2^8 - 1)$ or 255 comparators. Encoder logic would be more complex, requiring a 256-line-to-8-line priority encoder.

15-8 FREQUENCY RESPONSE OF ADCs

15-8.1 Aperture Error

During conversion time, T_C , the analog input voltage must not change by more than $\pm \frac{1}{2}$ LSB (total 1 LSB), or the conversion will be incorrect. This type of inaccuracy is called *aperture error*. The rate of change of V_{in} with respect to time is called *slew rate*. If V_{in} is a sine wave, its slew rate is maximum at its zero crossings. The sine wave's slew rate is determined by both its peak voltage and frequency.

For an A/D converter, the maximum frequency for a sine wave V_{in} to be digitized within an accuracy of $\pm \frac{1}{2}$ LSB is

$$f_{\max} \approx \frac{1}{2\pi (T_C) 2^n} \quad (15-11)$$

Example 15-7

The AD670 is an 8-bit ADC with a conversion time of $10 \mu\text{s}$. Find the maximum frequency of an input sine wave that can be digitized without aperture error.

Solution From Eq. (15-11),

$$f_{\max} \approx \frac{1}{2\pi (2^8) 10 \mu\text{s}} = \frac{1}{2\pi (256) 10 \times 10^{-6} \text{ s}} = 62 \text{ Hz}$$

Example 15-7 shows that the frequency response of even a fast ADC is surprisingly low. For a 10-bit integrating ADC with a conversion time of $\frac{1}{3} \text{ s}$, the highest sine frequency is about 0.5 mHz, or 1 cycle per 2000 s.

Summary. An 8-bit converter with a $10\text{-}\mu\text{s}$ conversion time can theoretically perform $[1/(10 \mu\text{s})]\mu\text{s} = 100,000$ conversions per second, yet the highest frequency sine wave that can be converted without slew-rate limiting is about 62 cycles per second. To raise the frequency response, we must add another circuit block, the sample-and-hold or follower-and-hold amplifier.

15-8.2 Sample-and-Hold Amplifier

The sample-and-hold (S/H) or follow-and-hold amplifier of Fig. 15-13 is made from two op amps, a hold capacitor (C_H), and a high-speed analog switch. This amplifier is connected between an analog input signal and the input to an ADC.

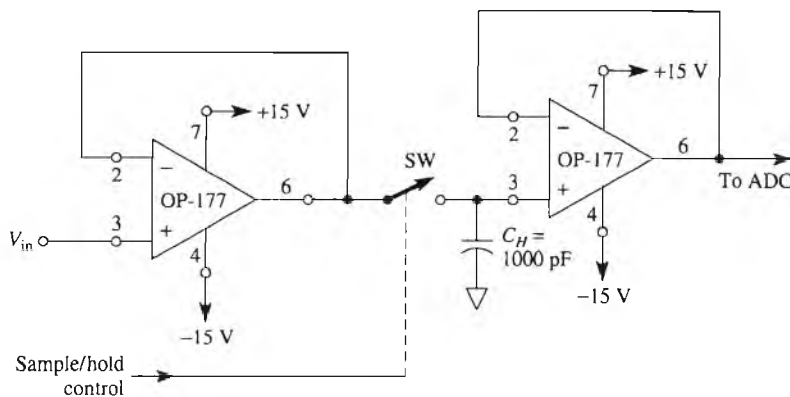


FIGURE 15-13 Sample-and-hold amplifier.

When the S/H amplifier is in the *sample* mode, the switch is closed and hold capacitor (C_H) voltage follows V_{in} . A *hold* command opens the switch and C_H retains a charge equal to V_{in} at the moment of switching. The S/H amplifier thus acts to hold V_{in} (stored on C_H) constant, while the ADC performs a conversion.

Conversion time of the ADC no longer limits frequency response. Instead, the limit is the *aperture time* of the S/H amplifier, which can be made much less than the conversion time. Aperture time is the time elapsed between a hold command and a switch opening. If the hold command is advanced by a time equal to the aperture time, C_H will hold the desired sample of V_{in} . Then the only remaining error is *aperture time uncertainty*, the switch jitter variation for each hold command.

Commercial S/H amplifiers have aperture time uncertainties lower than 50 ns. An example shows the improvement in frequency response due to an added S/H amplifier.

Example 15-8

An S/H amplifier with an aperture time uncertainty of 50 ns is connected to an 8-bit ADC. Find the highest-frequency sine wave that can be digitized within an error of 1 LSB.

Solution Replace conversion time by aperture uncertainty time in Eq. (15-11):

$$f_{\max} \approx \frac{1}{2\pi(2^8)50 \times 10^{-9} \text{ s}} = 12.4 \text{ kHz}$$

PROBLEMS

- 15-1. A 10-bit A/D converter is capable of accepting an input voltage 0 to 5.12 V. What is the digital output code if the input voltage is 2.4 V?

- 15-2. If the 8-bit A/D converter of Example 15-2 has an offset error of $-\frac{1}{2}$ LSB, what analog input voltage will cause the digital output to be all 1s?
- 15-3. Note that the A/D converter of Example 15-1 has an offset error of $+\frac{1}{2}$ LSB and a gain error of -0.4% of FSR. What is the analog input voltage needed to cause the digital output to be all 1s?
- 15-4. Name three types of ADCs and indicate their relative conversion speeds (slow or fast).
- 15-5. $V_{in} = 50$ mV in the integrating ADC circuit of Fig. 15-5.
(a) What is the duration for integrating phase T_1 and the value of V_o ?
(b) What is the name of phase T_2 , the value of V_{ref} , and the duration of T_2 ?
(c) Find the circuit output.
- 15-6. Name the three components of a successive approximation 8-bit ADC.
- 15-7. A microprocessor issues a write command to an ADC. Does the ADC send data to the microprocessor or perform a conversion?
- 15-8. An input voltage with a range of 0 to 2.55 V is applied to pin 16 of an AD670 and pin 18 is grounded. Which other input pins should be jumpered or grounded to select this range?
- 15-9. How do you pin-program the AD670 for straight binary output?
- 15-10. How does a microprocessor “tell” an AD670 to (a) perform a conversion; (b) place the result on the data bus? (c) How does the microprocessor know when the AD670 has finished a conversion and its data are valid?
- 15-11. (a) What is the conversion time for an AD670?
(b) How many conversions can it perform per second?
(c) What is the maximum sine wave frequency that it can convert without adding a sample-and-hold amplifier?
- 15-12. If the sample-and-hold amplifier of an 8-bit ADC has an aperture uncertainty time of 10 ns, what maximum sine wave frequency can it convert within $\pm\frac{1}{2}$ LSB?
- 15-13. How many comparators are required to make an 8-bit flash converter?